



4.4.1 A has linearly independent columns ¶

04.4.1 Solving LLS via QR factorization

fit width



Theorem 4.4.1.1. Assume $A \in \mathbb{C}^{m \times n}$ has linearly independent columns and let $A = QR$ be its QR factorization with orthonormal matrix $Q \in \mathbb{C}^{m \times n}$ and upper triangular matrix $R \in \mathbb{C}^{n \times n}$. Then the LLS problem

$$\text{Find } \hat{x} \in \mathbb{C}^n \text{ such that } \|b - A\hat{x}\|_2 = \min_{x \in \mathbb{C}^n} \|b - Ax\|_2$$

is solved by the unique solution of

$$R\hat{x} = Q^H b.$$

Proof 1.

Since $A = QR$, minimizing $\|b - Ax\|_2$ means minimizing

$$\|b - Q \underbrace{Rx}_z\|_2.$$

Since R is nonsingular, we can first find z that minimizes

$$\|b - Qz\|_2$$

after which we can solve $Rx = z$ for x . But from the Method of Normal Equations we know that the minimizing z solves

$$Q^H Qz = Q^H b.$$

Since Q has orthonormal columns, we thus deduce that

$$z = Q^H b.$$

Hence, the desired $\hat{x}$ must satisfy

$$R\hat{x} = Q^H b.$$

Proof 2.

Let $A = Q_L R_{TL}$ be the QR factorization of A . We know that then there exists a matrix Q_R such that $Q = \begin{pmatrix} Q_L & Q_R \end{pmatrix}$ is unitary: Q_R is an orthonormal basis for the space orthogonal to the space spanned by Q_L . Now,

$$\begin{aligned} & \min_{x \in \mathbb{C}^n} \|b - Ax\|_2^2 \\ &= < \text{substitute } A = Q_L R_{TL} > \\ & \min_{x \in \mathbb{C}^n} \|b - Q_L R_{TL} x\|_2^2 \\ &= < \text{two norm is preserved since } Q^H \text{ is unitary} > \\ & \min_{x \in \mathbb{C}^n} \|Q^H(b - Q_L R_{TL} x)\|_2^2 \\ &= < \text{partitioning; distributing} > \\ & \min_{x \in \mathbb{C}^n} \left\| \begin{pmatrix} Q_L^H \\ Q_R^H \end{pmatrix} b - \begin{pmatrix} Q_L^H \\ Q_R^H \end{pmatrix} Q_L R_{TL} x \right\|_2^2 \\ &= < \text{partitioned matrix-matrix multiplication} > \\ & \min_{x \in \mathbb{C}^n} \left\| \begin{pmatrix} Q_L^H b \\ Q_R^H b \end{pmatrix} - \begin{pmatrix} R_{TL} x \\ 0 \end{pmatrix} \right\|_2^2 \\ &= < \text{partitioned matrix addition} > \\ & \min_{x \in \mathbb{C}^n} \left\| \begin{pmatrix} Q_L^H b - R_{TL} x \\ Q_R^H b \end{pmatrix} \right\|_2^2 \\ &= < \text{property of the 2-norm: } \left\| \begin{pmatrix} u \\ v \end{pmatrix} \right\|_2^2 = \|u\|_2^2 + \|v\|_2^2 > \\ & \min_{x \in \mathbb{C}^n} \left(\|Q_L^H b - R_{TL} x\|_2^2 + \|Q_R^H b\|_2^2 \right) \\ &= < Q_R^H b \text{ is independent of } x > \\ & \left(\min_{x \in \mathbb{C}^n} \|Q_L^H b - R_{TL} x\|_2^2 \right) + \|Q_R^H b\|_2^2 \\ &= < \text{minimized by } \hat{x} \text{ that satisfies } R_{TL} \hat{x} = Q_L^H b > \\ & \|Q_R^H b\|_2^2. \end{aligned}$$

Thus, the desired $\hat{x}$ that minimizes the linear least squares problem solves $R_{TL}\hat{x} = Q_L^H y$. The solution is unique because R_{TL} is nonsingular (because A has linearly independent columns).

Homework 4.4.1.1. Yet another alternative proof for [Theorem 4.4.1.1](#) starts with the observation that the solution is given by $\hat{x} = (A^H A)^{-1} A^H b$ and then substitutes in $A = QR$. Give a proof that builds on this insight.

▼ [Solution](#)

Recall that we saw in [Subsection 4.2.2](#) that, if A has linearly independent columns, the LLS solution is given by $\hat{x} = (A^H A)^{-1} A^H b$ (the solution to the normal equations). Also, if A has linearly independent columns and $A = QR$ is its QR factorization, then the upper triangular matrix R is nonsingular (and hence has no zeroes on its diagonal).

Now,

$$\begin{aligned} \hat{x} &= \text{< Solution to the Normal Equations >} \\ &= (A^H A)^{-1} A^H b \\ &= \text{< } A = QR \text{ >} \\ &= [(QR)^H (QR)]^{-1} (QR)^H b \\ &= \text{< } (BC)^H = (C^H B^H) \text{ >} \\ &= [R^H Q^H Q R]^{-1} R^H Q^H b \\ &= \text{< } Q^H Q = I \text{ >} \\ &= [R^H R]^{-1} R^H Q^H b \\ &= \text{< } (BC)^{-1} = C^{-1} B^{-1} \text{ >} \\ &= R^{-1} R^{-H} R^H Q^H b \\ &= \text{< } R^{-H} R^H = I \text{ >} \\ &= R^{-1} Q^H b. \end{aligned}$$

Thus, the $\hat{x}$ that solves $R\hat{x} = Q^H b$ solves the LLS problem.

Ponder This 4.4.1.2. Create a picture similar to [Figure 4.3.2.1](#) that uses the QR factorization rather than the SVD.